SNNTracker: Online High-Speed Multi-Object Tracking With Spike Camera

Yajing Zheng, *Associate Member, IEEE*, Chengen Li, Jiyuan Zhang, *Member, IEEE*, Zhaofei Yu, *Member, IEEE*, and Tiejun Huang, *Senior Member, IEEE*

Abstract*—Multi-object tracking (MOT) is crucial for applications such as autonomous driving and robotics, yet traditional image-based methods struggle in high-speed scenarios due to motion blur and temporal gaps caused by low frame rates. Spike cameras, with their ability to continuously record spatiotemporal signals, overcome these limitations. However, existing spike-based methods often rely on intermediate image reconstruction or discrete clustering, limiting real-time performance and temporal continuity. To address this, we propose SNNTracker, the first fully spiking neural network (SNN)-based MOT algorithm tailored for spike cameras. SNNTracker integrates a dynamic neural field (DNF)-based attention mechanism for target detection and a winner-take-all (WTA)-based tracking module with online spike-timing-dependent plasticity (STDP) for adaptive learning of object trajectories. By directly processing spike streams without reconstruction, SNNTracker reduces latency, computational overhead, and dependency on image quality, making it ideal for ultra-high-speed environments. It maintains robust, continuous tracking even under occlusions, severe lighting variations, or temporary object disappearance, by leveraging SNN-estimated motion predictions and long-term online clustering. We construct three types of spike-camera MOT datasets covering dense and sparse annotations across diverse real-world scenarios, including camera ego-motion, deformable and ultra-fast motion (up to 2600 RPM), occlusion, indoor/outdoor lighting changes, and low-visibility tracking. Extensive experiments demonstrate that SNNTracker consistently outperforms state-of-the-art MOT methods—both ANN- and SNN-based—achieving MOTA scores above 96% and up to 100% in many sequences. Our results highlight the advantages of spike-driven SNNs for low-latency, high-speed, and label-free multi-object tracking, advancing neuromorphic vision for real-time perception.

***Index Terms*—DNF, high-speed MOT, online learning, STDP, spike cameras, spiking neural network, WTA.**

Received 20 December 2024; revised 16 July 2025; accepted 9 September 2025. Date of publication 16 September 2025; date of current version 3 December 2025. This work was supported in part by the Young Elite Scientists Sponsorship Program by CAST under Grant 2023QNRC001, in part by Beijing Natural Science Foundation under Grant 4252039, in part by the National Natural Science Foundation of China under Grant 62306015, and in part by the China Postdoctoral Science Foundation under Grant 2023T160015. Recommended for acceptance by J. Gu. *(Corresponding author: Yajing Zheng.)*

Yajing Zheng and Jiyuan Zhang are with the State Key Laboratory for Multimedia Information Processing, School of Computer Science, Peking University, Beijing 100871, China (e-mail: yj.zheng@pku.edu.cn; jyzhang@stu.pku.edu.cn).

Chengen Li is with the School of Software and Microelectronics, Peking University, Beijing 100871, China (e-mail: 2201210734@stu.pku.edu.cn).

Zhaofei Yu and Tiejun Huang are with the Institute for Artificial Intelligence, Peking University, Beijing 100871, China, and also with the State Key Laboratory for Multimedia Information Processing, School of Computer Science, Peking University, Beijing 100871, China (e-mail: yuzf12@pku.edu.cn; tjhuang@pku.edu.cn).

Digital Object Identifier 10.1109/TPAMI.2025.3610696

I. Introduction

TRACKING multiple high-speed moving objects is a fundamental and urgent problem in computer vision, with applications spanning autonomous driving, surveillance, and uncrewed aerial vehicles (UAV). In high-speed dynamic environments, the rapid movement of objects or cameras often introduces motion blur, making it difficult to estimate object motion and location accurately. Increasing the camera's shutter speed can mitigate motion blur; however, it results in exponentially increasing computational demands and data volumes. For example, the current fastest tracking approach [1] achieves single-object tracking at 200 fps but suffers from reduced accuracy. Scaling such methods to handle multiple objects becomes impractical when millions or even hundreds of millions of images are generated by high-speed digital cameras every second [2], [3].

In recent years, event cameras have emerged as a promising alternative to address the challenges of high-speed tracking [4]. Unlike traditional cameras, which capture scenes with a fixed exposure time for all pixels, neuromorphic vision sensors asynchronously generate binary events based on changes in scene radiance. Event cameras, in particular, produce events only when the brightness change exceeds a specific threshold [5], [6], [7]. This approach offers significant advantages, such as low latency, high dynamic range (HDR), and reduced data redundancy, making it highly suitable for high-speed detection and tracking tasks [7], [8], [9], [10], [11], [12]. While event cameras effectively manage data bandwidth and high-speed acquisition through event differencing, their highly discrete nature introduces challenges. Few algorithms can truly achieve event-driven processing; instead, most methods aggregate events into voxels or frames before feeding them into networks for processing [7], [13], [14], [15], [16]. This limitation has given rise to an active area of research focusing on how to effectively represent event streams in ways that fully leverage their asynchronous and sparse nature.

In parallel, spike cameras [17], [18], another type of neuromorphic vision sensor, have shown remarkable potential for recording spatiotemporal signals continuously in ultra-high-speed and HDR conditions. Unlike conventional cameras, spike cameras generate spike streams directly in response to photon accumulation, enabling continuous visual signal capture with minimal latency. This unique capability makes spike cameras highly advantageous for tracking multiple fast-moving objects in real time. However, existing spike-based tracking

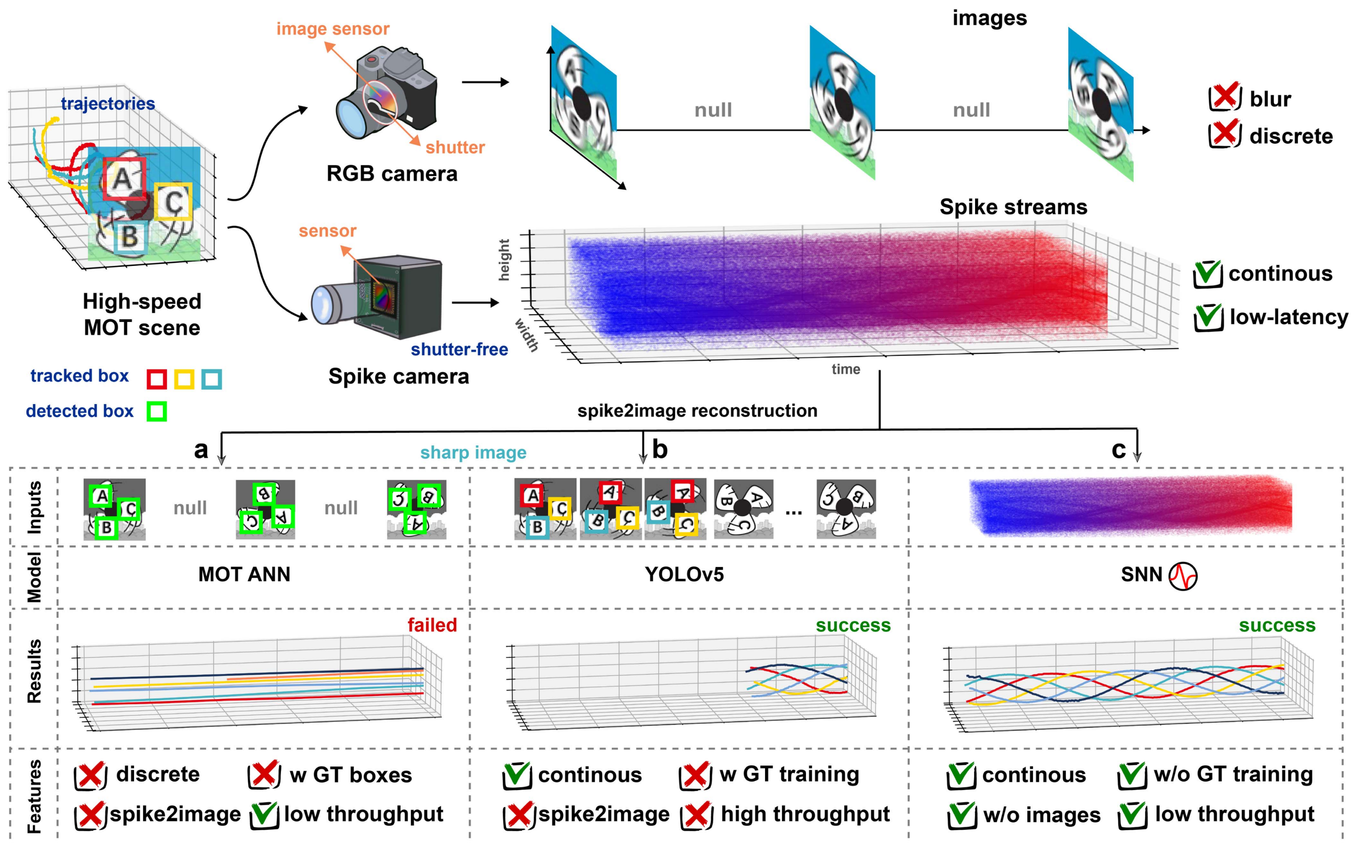

Fig. 1. Comparison between traditional RGB cameras and spike cameras for high-speed multi-object tracking tasks. Traditional RGB cameras capture discrete images that often suffer from motion blur, whereas spike cameras generate continuous, low-latency spike streams. (a) While spike cameras enable the reconstruction of frame-skipped, sharp images combined with detection boxes, it remains challenging for MOT algorithms to associate and track high-speed objects effectively. (b) Continuous reconstruction of sharp images with labeled tracking boxes allows YOLOv5 to perform well on annotated validation datasets. However, this method is heavily dependent on labeled data and introduces significant data overhead compared to utilizing raw spike streams. (c) SNN can direct tracking on raw spike streams, eliminating the need for labeled data, and achieves accurate, real-time tracking of multiple ultra-fast-moving objects with minimal latency.

frameworks, such as the SVS (Spike Vision System) [17], [18] and ODTSNet [19], are still rooted in detection-based MOT paradigms [20]. Specifically, SVS applies SNN-based filters to process spike streams and detect motion-induced spikes, followed by spatial association using the Hungarian algorithm [21] and Kalman filtering [22]. ODTSNet improves upon SVS by incorporating STDP [23], [24]-based motion estimation and DBSCAN [25] clustering to separate object motion from camera ego-motion. Despite these advancements, these methods suffer from two main limitations: (1) their reliance on intermediate spike stream reconstruction or parameter-sensitive clustering algorithms introduces computational overhead and delays, and (2) the tracking accuracy deteriorates under dense spike streams or complex high-speed conditions.

Why Spike? This question has been a focal point of discussion among computational neuroscience researchers and various forums as spike-based computation continues to gain momentum [26], [27], [28], [29], [30], [31]. The benefits of spike processing are particularly evident in its ability to handle asynchronous and event-driven data with minimal latency and energy consumption—two properties that align perfectly with the demands of high-speed, real-time tasks. Traditional RGB cameras, as illustrated in Fig. 1, face significant challenges in ultra-high-speed environments due to motion blur and frame discontinuities. By contrast, spike cameras generate continuous and low-latency spike streams, which are highly suitable for seamless multi-object tracking. Although sharp images can be reconstructed from continuous spike streams and state-of-the-art MOT algorithms can be applied, the performance remains constrained by the quality of the reconstructed images and the additional latency introduced by the reconstruction process. Moreover, discrete sharp images still struggle with object association in scenarios involving ultra-fast motion. While methods such as YOLOv5 [32] trained on continuous reconstructed images combined with labeled detection boxes achieve reasonable tracking performance, they suffer from increased data volume and dependence on annotated datasets.

To address these challenges, we propose the first fully SNN-based multi-object tracking algorithm tailored for spike cameras, *SNNTracker*. Unlike existing detection-based frameworks, SNNTracker directly processes spike streams without reconstruction, significantly reducing latency and computational overhead. Our method introduces two key innovations: (1) a dynamic neural field-based attention mechanism that simulates visual saccades for efficient top-down target detection, and (2) a WTA [33]-based tracking module that employs STDP for online learning of object trajectories. By integrating motion predictions estimated by SNNs, SNNTracker ensures robust and continuous tracking, even under challenging scenarios such as occlusions, temporary target disappearance, and camera ego-motion.

To validate our approach, we construct new spike camera MOT datasets featuring diverse and complex tracking scenarios, including high-speed rotating objects, varying illumination conditions, and continuous annotations for camera motion and object occlusions. Additionally, we upgrade an existing spike-based detection dataset by introducing temporal associations to evaluate the impact of temporal continuity on MOT performance. Through extensive experiments, we demonstrate that SNNTracker consistently achieves superior performance compared to MOT SOTA [34], [35], [36], [37], [38], including supervised approaches such as YOLOv5 [32], [39]. Notably, our method achieves MOTA scores exceeding 96% across all datasets, with several sequences reaching 100%, showcasing the robustness and efficiency of our approach.

In summary, our contributions are as follows:

- We propose SNNTracker, the first fully SNN-based MOT algorithm for spike cameras, which integrates top-down attention and STDP-driven online learning to enable real-time, low-latency tracking without requiring labeled data.
- We construct diverse spike-camera MOT datasets with varying annotation granularities, covering indoor/outdoor scenes, occlusions, lighting changes, and continuous high-speed motion.
- We conduct in-depth comparative analyses to highlight the impact of temporal continuity and motion blur on MOT performance, demonstrating the advantages of SNNs for spike stream processing in ultra-high-speed environments.

The rest of this paper is organized as follows: Section II reviews related work on MOT algorithms and spike camera applications. Section III details the proposed SNNTracker framework. Section IV presents experimental results and comparisons. Finally, Section V concludes the paper and discusses future directions.

II. Related Works

A. Spike Camera

Spike cameras are a novel type of neuromorphic vision sensor inspired by the phototransduction mechanism of the primate fovea [17], [18]. Unlike traditional cameras that rely on fixed exposure times and global shutter operations, spike cameras asynchronously accumulate light intensity at each pixel. When the accumulated voltage exceeds a threshold, a spike is emitted, and the spike's temporal distribution encodes the pixel's brightness. This design eliminates the trade-off between shutter speed and exposure time, enabling ultra-high-speed and high-dynamic-range (HDR) visual capture without motion blur or temporal gaps [40]. Recent studies have demonstrated the immense potential of spike cameras in tasks such as high-speed imaging [41], [42], [43], [44], [45], HDR reconstruction [46], and 3D scene synthesis [47]. Beyond imaging, spike cameras excel in high-speed detection [48], tracking [19], and recognition tasks [49], leveraging the continuity and low-latency characteristics of spike streams. For instance, spike-guided image enhancement [50], [51], [52] has been explored by combining spike cameras with traditional RGB cameras, where spike data aids in deblurring and restoring motion-blurred RGB images. However, despite advances in reconstruction-based methods, these approaches often suffer from latency and increased data volume. In contrast, directly processing spike streams for tasks such as multi-object detection and tracking [17], [19] offers significant advantages in latency reduction, computational efficiency, and robustness to high-speed motion. These capabilities make spike cameras a promising tool for real-time visual analysis in ultra-dynamic environments.

B. Multiple Object Tracking

MOT has long been a central task in computer vision, with applications ranging from surveillance and autonomous driving to robotics. Most MOT methods follow a framework for detection and data association, where object detectors, such as YOLOv5 [32], generate boundary boxes, and association mechanisms link detections across frames to form continuous trajectories. Two primary association strategies include IoU-based matching, which relies on spatial overlap between detections, and appearance-based matching through Re-Identification (ReID) features [53], [54], which use visual cues to maintain object identities under occlusions or long-term absence. ReID has become a cornerstone of MOT algorithms, enabling robust handling of identity switches and fragmented tracks. Classical methods like DeepSORT [36] combine motion models (e.g., Kalman filtering) with deep appearance embeddings to achieve reliable data association. Recent works, such as ByteTrack [38], further optimize this pipeline by retaining low-confidence detections, which are often filtered out prematurely, to enhance object continuity. Transformer-based models [55] and graph-based approaches [56], [57] have also been introduced to incorporate global feature aggregation, improving tracking accuracy in complex scenarios.

However, these SOTA algorithms face significant limitations in high-speed environments. Rapid object or camera movement induces motion blur, which severely degrades detection quality, leading to missed or imprecise bounding boxes. This directly affects ReID features, which rely heavily on sharp and distinct object appearances. Furthermore, traditional cameras with low frame rates produce discrete image sequences, resulting in temporal gaps that disrupt motion continuity and increase the risk of identity switches. These challenges expose the inadequacy of existing MOT frameworks in ultra-high-speed scenarios, where precise and low-latency tracking is critical.

While event cameras [7], [58], [59] have also been widely studied in neuromorphic vision, we do not adopt them in our system due to their limited ability to provide temporally continuous and dense representations, especially in scenarios with subtle brightness variations or low-contrast motion. Unlike event cameras, which only emit signals upon brightness changes, spike cameras produce a more stable and temporally continuous stream, which is particularly beneficial for robust tracking. More

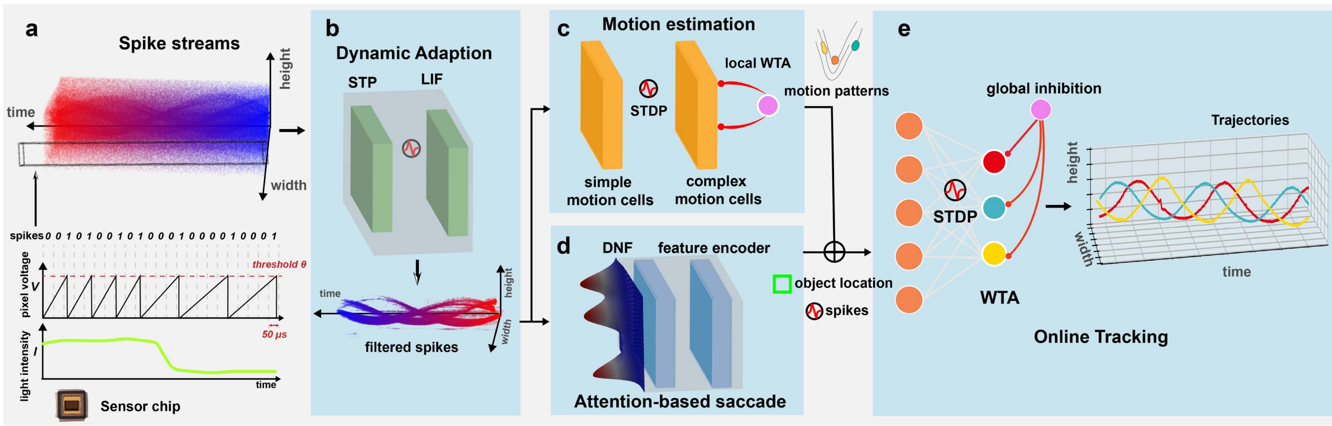

Fig. 2. Illustration of the proposed fully SNN-based MOT framework, SNNTracker. (a) Spike camera principles and the spike stream input to the network. (b) Dynamic adaptation filters out motion-irrelevant spikes from the spike streams. (c) Online motion estimation using STDP identifies motion patterns present in the filtered spikes. (d) The DNF-based attention saccade module locates different moving objects and encodes their features into spikes. (e) The WTA-based online tracking module adjusts the weights of output neurons dynamically using the STDP rule to track different targets.

detailed comparisons, theoretical analysis, and experimental evidence supporting the superiority of spike cameras in continuous high-speed tracking tasks are provided in the supplementary material.

III. Methodology

In this section, we introduce SNNTracker, a multi-layer spiking neural network designed for high-speed object tracking. The architecture of SNNTracker is illustrated in Fig. 2. The network comprises four main components: adaptation, motion estimation, attention-based saccade, and WTA-based tracking. By directly processing spike streams, SNNTracker enables the continuous tracking of each moving object, accurately capturing their trajectories in real time. The following subsections detail the principles and functionalities of each component.

A. Dynamic Adaption

1) Spike Camera Mechanism: Spike cameras eliminate the traditional shutter mechanism and instead use standard CMOS technology to continuously sense external light intensity I. Once powered on, the camera performs uninterrupted photoelectric conversion. Within the spike pixel array, each pixel accumulates the light intensity signal I it receives over time. The accumulated value $V(t)$ at each pixel evolves as:

$$V(t) = V(t - \Delta t) + I(t) \cdot \Delta t, \tag{1}$$

where Δt represents the integration time step. When $V(t)$ exceeds a predefined threshold θ, the pixel generates a spike and resets $V(t)$ to its baseline value:

$$\text{If } V(t) \geq \theta, \text{ generate spike and reset } V(t) \to 0. \tag{2}$$

This mechanism ensures that the spike generation process is continuous and responsive to light intensity variations.

Currently, several spike camera models are available, with maximum output frequencies reaching up to 40 kHz. In this study, we use the *Spike Camera-001T-Gen2* [18], which operates at an output frequency of 20 kHz. The camera has a spatial resolution of 400 (H) $\times$ 250 (W) pixels and a dynamic range of $\geq$110 dB. As shown in Fig. 2(a), the camera's photoreceptive chip reads each pixel at 50-μs intervals to determine whether a spike has been generated. If a spike is detected, the output is set to 1; otherwise, it is set to 0. This process produces a binary spike signal S for each pixel, forming a spike stream with dimensions:

$$S \in \{0, 1\}^{H \times W \times T}, \tag{3}$$

where H and W represent the spatial resolution of the pixel array, and T denotes the temporal steps.

The spike generation mechanism ensures that pixels exposed to higher light intensities generate spikes at a higher frequency, making the spike frequency f_s approximately proportional to the light intensity $I(t)$ as:

$$f_s \propto \frac{I(t)}{\theta}. \tag{4}$$

From the spike generation principle, it is evident that pixels exposed to stronger light intensities fire spikes at higher frequencies. This feature is particularly advantageous for reconstructing high-speed video. However, when analyzing object motion based on the spike camera's output, redundant spikes from static pixels often interfere with subsequent high-level visual tasks.

2) Temporal Filtering With STP: From the principle of spike generation, it is evident that motion induces changes in light intensity at the corresponding pixel locations. Since the spike generation frequency is proportional to the light intensity, it is feasible to determine whether a pixel is associated with motion through a filter applied to the spike input. Following the approaches in SVS [17] and ODTSnet [19], we adopt a temporal filtering mechanism based on Short-Term Plasticity (STP) to process spike streams in the time domain.

The spike camera outputs, represented as 3-element tuples $\{x, y, t\}$, are first passed through the temporal filter. The postsynaptic potential of each neuron, which corresponds to the pixel locations in the spike stream, is modulated by the temporal regularity of the input spikes $\mathcal{S}_{ij}$. For input spikes with a fixed frequency, the corresponding postsynaptic potential converges to a steady state after several spikes. Specifically, we use a short-term-facilitation-dominated model [60] to detect motion by examining the difference between R_n and R_{n+1}, where n is the spike index. The dynamics of R and u are governed by the following equations:

$$R_{n+1} = 1 - [1 - R_n(1 - u_n)] \exp\left(-\frac{\Delta t_n}{\tau_D}\right), \tag{5}$$

$$u_{n+1} = U + [u_n + C(1 - u_n) - U] \exp\left(-\frac{\Delta t_n}{\tau_F}\right), \tag{6}$$

where R_n and u_n denote the values of R and u between spikes n and $n+1$, and Δt_n is the time interval between these spikes. The parameters τ_D and τ_F represent the decay and facilitation time constants, respectively, while U and C are constants defining the dynamics.

The postsynaptic spike is triggered based on the change in R, governed by the following condition:

$$\mathcal{I}_{ij} = \begin{cases} 1, & |R_{n+1} - R_n| \geq \vartheta, \\ 0, & |R_{n+1} - R_n| < \vartheta, \end{cases} \tag{7}$$

where $\mathcal{I}_{ij}$ is a binary indicator denoting whether the postsynaptic neuron fires (1) or not (0), determined by whether the absolute change in R exceeds a predefined threshold ϑ.

This update rule ensures that the temporal filter operates entirely in a spike-driven manner, only activating when spikes are present. Such an approach leverages the energy efficiency inherent to spike-based computation. In the Section. IV, we will further discuss the advantages of this implementation in hardware, specifically its performance on FPGA platforms.

3) Spatial Filtering: A spatial filtering mechanism is introduced to suppress noise further. This spatial filtering is implicitly achieved by using the filtered spikes $\mathcal{I}$ obtained from the temporal filter layer as afferent inputs to leaky integrate-and-fire (LIF) neurons [61]. The sub-threshold dynamics of an LIF neuron are described by:

$$\tau_m \frac{dv(t)}{dt} = -[v(t) - v_{rest}] + R\mathbb{I}(t), \tag{8}$$

where $v(t)$ represents the membrane potential of the LIF neuron at time t, τ_m is the membrane time constant, v_{rest} denotes the resting potential, and R is the resistance. The input current $\mathbb{I}(t)$ is computed as the sum of afferent spikes from the eight-connected neighboring neurons in the temporal filter layer:

$$\mathbb{I}(t) = \sum_{\boldsymbol{x}} \mathcal{I}_{\boldsymbol{x}},$$

where $\boldsymbol{x}$ denotes the position of a pre-layer neuron connected to the LIF neuron.

Upon receiving the integrated spike input, the LIF neuron updates its state according to Eq. (8). When the membrane potential $v(t)$ exceeds a predefined threshold Θ, the neuron fires a spike and resets its potential to the resting value v_{rest} as follows:

$$\lim_{\delta \to 0^+} v(t + \delta) = \begin{cases} v_{rest}, & \text{if } v(t) \geq \Theta, \\ v(t), & \text{if } v(t) < \Theta. \end{cases} \tag{9}$$

This process ensures that the LIF neuron only activates when the integrated membrane potential surpasses the threshold Θ. Because the LIF neurons are locally connected to their pre-layer counterparts and operate exclusively when the integrated voltage exceeds the threshold, random noise events/spikes are effectively reduced. Consequently, the spatial filtering process enhances the robustness of the system by selectively responding to meaningful input signals while suppressing irrelevant noise.

B. Motion Estimation

1) Simple Motion Cells: In this layer, we define 8×4 neurons at each pixel, corresponding to eight motion directions and four motion speeds. The motion pattern $\boldsymbol{m}$ of each spike is determined by a weighted sum of the responses from these motion neurons:

$$\boldsymbol{m} = \sum_{k=1}^{32} w^k \boldsymbol{v}^k, \tag{10}$$

where w^k represents the synaptic weight, and $\boldsymbol{v}^k$ denotes the motion vector of the k-th motion neuron. This formulation captures the overall motion pattern at each pixel based on the contributions of all 32 neurons.

2) Online Motion Learning: At each pixel location, the 32 synaptic weights are dynamically updated to adapt to the motion pattern of the pixel. To achieve this, we propose an unsupervised learning algorithm based on the STDP (Spike-Timing Dependent Plasticity) learning rule. The synaptic weights of the motion neurons are adjusted by comparing the spike timings between postsynaptic spikes at timestamp $t+1$ and filtered presynaptic spikes in the input layer at timestamp t. Specifically, the synaptic weight w_i^k of the k-th motion neuron at pixel i (simplified to one dimension for clarity) is updated using the following online STDP rule:

$$\Delta w_i^k = \eta \frac{\kappa * \boldsymbol{A_i}}{N_i}, \tag{11}$$

$$\boldsymbol{A_i} = A_+ \Gamma(t_{k,j}^{pre} - t_{k,j}^{post}) - A_- |t_{k,j}^{pre} - t_{k,j}^{post}|, \tag{12}$$

where η is the learning rate, and j refers to the neighboring neurons of neuron i. The term N_i represents the number of firing spikes in the neighborhood of neuron i, calculated as:

$$N_i = \sum_{j=1}^{j \in Nei(i)} \Gamma(t_{k,j}^{post} - t). \tag{13}$$

The function $\Gamma(x)$ is an indicator that equals 1 only when $x = 0$. Parameters A_+ and A_- specify the magnitude of change associated with long-term potentiation and long-term depression, respectively. The term κ is a convolutional kernel with a two-dimensional Gaussian distribution and a mean of $\mu = 1$.

3) Complex Motion Cells: In regions with dense spikes, the motion learning of neurons may be influenced by neighboring spikes, potentially causing different motion cells to acquire identical synaptic weights. This phenomenon, referred to as *motion confusion*, is illustrated in the supplementary material.

To mitigate motion confusion and limit the number of motion patterns within a local region, we introduce complex motion cells equipped with local WTA circuits. These circuits ensure that the most frequent motion vector dominates the motion pattern of the region. After extracting motion patterns using the rectified STDP learning rule, lateral competition among motion neurons is introduced over a region larger than the neighborhood size defined in Eqs. 11 and 12. If multiple neurons emerge as winners, all motion cells in the region are suppressed to prevent over-representation of motion patterns.

This mechanism not only addresses motion confusion but also alleviates the aperture problem [62], as considering a larger region in the WTA competition naturally reduces ambiguities in motion detection.

C. Attention-Based Saccade

This module that mimicks the saccade process of eye [63], [64] and "looks" the moving objects sequentially.

1) DNF-Based Attention: When multiple moving objects are present in the field of view, it is naturally impossible to observe them all clearly at the same time. Instead, we focus on these objects sequentially, following a certain priority order. For instance, when observing a human face, we typically focus on the eyes first, followed by the nose, ears, and other facial features. This process involves shifting attention to different parts of the scene through eye movements, known as *saccades*, to understand the external environment.

In computer vision, object detection and tracking tasks often draw inspiration from attention mechanisms. Examples include the *transformer* [65], [66] and the *region proposal network (RPN)* used in Faster R-CNN [67]. However, these methods primarily rely on features extracted from image data through convolutional blocks in deep neural networks, making them unsuitable for processing spike streams generated by spiking cameras.

To effectively cluster spikes corresponding to each object, we leverage a DNF [68] to identify attractors associated with different objects. We then selectively attend to the spikes located in these attractor regions. In DNF, the dynamics of the neural system are expressed as a space-time continuous distribution of neural activation $u(z,t)$, where z denotes the spatial location and t represents the time-step. The spatial location z lies in a two-dimensional space matching the size of the filtered spikes obtained from the *Dynamic Adaptation* module. For simplicity, we use a one-dimensional description in the following discussion.

The neural activation $u(z,t)$ evolves according to the following nonlinear dynamics [69]:

$$\tau\dot{u}(z,t) = -u(z,t) + \int_{z'} J(z,z')r(z',t)dz' + I^{ext}(z,t), \tag{14}$$

where τ is the synaptic time constant, $I^{ext}(z,t)$ is the filtered input from the dynamic adaptation module, and $r(z',t)$ is the activation function of the neuron at location z' and time t. The interaction $J(z,z')$ describes the relationship between neurons at locations z and z', modeled as:

$$J(z,z') = \frac{J_0}{\sqrt{2\pi}a}\exp\left(-\frac{(z-z')^2}{2a^2}\right), \tag{15}$$

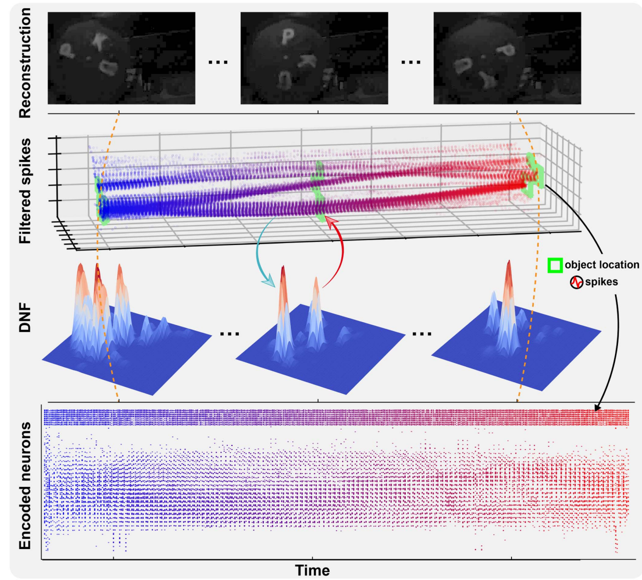

Fig. 3. Illustration of the attention-based saccade module that based on the DNF.

where J_0 is the interaction strength, and a controls the spread of the interaction.

The activation function $r(z,t)$ is determined by the input spikes and is normalized across the spatial domain:

$$r(z,t) = \frac{U_+(z,t)^2}{\int_{z'} U_+(z',t)^2 dz'}, \tag{16}$$

where $U_+(z,t) = U(z,t)$ if $U(z,t) > 0$, and $U_+(z,t) = 0$ otherwise.

2) Spike Feature Encoder: After updating the neural activation $u(\boldsymbol{z})$, the attractors (self-stabilized peaks) in the neural field represent regions with strong input, corresponding to the locations of moving objects. Therefore, the moving objects can be identified by locating the attractors within the neural field.

As illustrated in Fig. 3, the positions of the attractors are marked with detection boxes, and the corresponding filtered spike arrays within these regions are cropped. These cropped spike arrays are then flattened and used as input for the subsequent clustering network. Additionally, the locations and sizes of the selected regions are encoded as spike representations, serving as auxiliary information during tracking. In Fig. 3, the encoded neurons at the top of the neural field primarily represent this information. The encoded spike representations, along with the detection boxes, are subsequently forwarded to the tracking module for further processing.

D. Online Tracking

Unlike existing object detection approaches [1], [70], [71], [72], [73] that using supervised classification networks to separate each object and then associate the object in different moments, we propose an online object tracking pipeline, in which each output neurons of SNN will automatically "follow" the moving object.

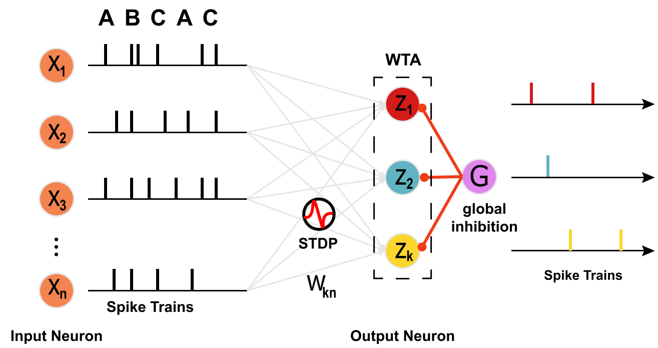

Fig. 4. Structure of the spike-based WTA circuit for object clustering. The number of input neurons matches the number of encoding neurons in the Attention-based Saccade module, receiving spikes generated by it. The output neurons correspond to the number of clusters. The connection weights between input and output neurons are updated online using the STDP rule. The spiking activity of output neurons indicates the category of the current input spikes, enabling clustering of different objects.

1) Tracking by Clustering: After obtaining the selected spike arrays via the attention selection module, we construct a spike-based WTA circuit to cluster the input spikes. The selected spike arrays will be fed into the circuit one-by-one, and output neurons will be trained to represent different clusters. Our clustering module consists of spiking neurons arranged in a WTA circuit that was inspired by the architecture of [74], [75]. An overview is shown in Fig. 4. The output neurons z_k are modeled by simplified Spike Response Model (SRM) [76], whose membrane potential is computed as the difference between the excitatory input $u_k(t)$ and the global inhibition term $G(t)$. $u_k(t)$ is the sum of excitatory inputs from neurons $x_1, \ldots, x_n$,

$$u_k(t) = w_{k0} + \sum_{i=1}^{n} w_{ki} \cdot x_i(t), \tag{17}$$

where w_{ki} is the synaptic weight of the clustering model, and the w_{k0} is the bias. In our cases, the $x_1, \ldots, x_n$ are the output of the attention selection modules. Similar to [75], we use a stochastic firing model for output neuron z_k, where the firing probability depends exponentially on the membrane potential $u_k(t)$,

$$p(z_k \text{ fires at time } t) \propto \exp(u_k(t) - G(t)). \tag{18}$$

Subsequently, we can model the firing behavior of each neuron z_k as an independent inhomogeneous Poisson process, whose instantaneous firing rate is given by,

$$r_k(t) = \exp u_k(t) - G(t). \tag{19}$$

In this case, if one of the neurons $z_1, \ldots, z_k$ fires an spike at time t, the conditional probability $q_k(t)$ that this spike originated from neuron z_k is:

$$q_k(t) = \frac{r_k(t)\delta t}{R(t)\delta t} = \frac{e^{u_k(t)}}{\sum_{i=1}^{K} e^{u_i(t)}} \tag{20}$$

where $R(t) = \sum_{i=1}^{K} r_k(t)$ is rate of an inhomogeneous Poisson process, which equals to the sum of the K independent Poisson process.

The synaptic weight between input neurons x_i and output neurons z_k in the clustering circuit change according to the STDP rule, which is trigger by spike of neuron z_k at time t. The specific update rule of the synaptic weight w_{ki} is:

$$\Delta w_{ki} = \begin{cases} e^{-w_{ki}} - 1, & x_i(t) = 1 \\ -1, & x_i(t) = 0 \end{cases}, \tag{21}$$

$$w_{ki}^t = w_{ki}^{t-1} + \Delta w_{ki}. \tag{22}$$

where yield a long-term potentiation (LTP) when the presynaptic neuron x_i fires before the postsynaptic neuron z_k, otherwise a long-term depression (LTD). In order to explore a form of intrinsic plasticity of the neurons, the bias weight w_{k0} of neuron z_k is updated as:

$$\Delta w_{k0} = e^{-w_{k0}} z_k - 1,$$
$$w_{k0}^t = w_{k0}^{t-1} + \eta \Delta w_{k0}. \tag{23}$$

The bias weight increases by inversely exponentially dependent on its current value when the neuron z_k fires, otherwise the bias weight is decreased by a constant.

2) Predictive Filtering: In the proposed clustering circuit, model weights are adjusted online based on the output of the attention selection module, ensuring that each output neuron z_k consistently corresponds to the same category. Different moving targets are assigned to distinct classes within the clustering network, allowing the output neurons to represent different targets and establish object associations across timestamps. Effectively, each output neuron z_k acts as a tracker, with its ID directly corresponding to k. To address ID switching and error detection in multi-object tracking, the motion estimation and attention selection modules are used to filter and validate the predicted IDs from the clustering network. Each tracker maintains attributes such as its ID, the bounding box of the associated target, movement speed, and predicted bounding box. The motion estimation module refines and complements tracking results, effectively replacing the predictive coding functionality traditionally performed by Kalman filtering [77], [78], [79], [80].

The bounding box of each tracker k is determined by the input corresponding to the neuron z_k in the clustering network, represented by the position and size of the spike array output from the attention selection module ($\mathbb{B}_k^t = \{x, y, h, w\}$). The object's moving speed $\boldsymbol{m}_k^t$ is calculated as the mean of the motion vectors within the region identified in the motion estimation results. Using the current position and moving speed, the predicted bounding box is computed as:

$$\hat{\mathbb{B}}^{t+1} = \mathbb{B}^t + \widetilde{\boldsymbol{m}}^t, \tag{24}$$

where $\widetilde{\boldsymbol{m}}^t = \{\boldsymbol{m}^t, 0, 0\}$ represents the augmented moving speed. These tracker attributes are then used to correct the detection results, with specific corrections applied depending on the scenario.

ID switch: If there are multiple objects with similar shapes, the trackers may confuse these objects during clustering, causing the IDs of different objects to exchange. Therefore, after obtaining the detection results of the clustering network, we first compare the IOU between the predicting bounding box $\hat{\mathbb{B}}_k^{t-1}$ of the tracker and its detected bounding box $\mathbb{B}_k^t$. If the IoU is less than 0.6, the current detection result is considered to be incorrect and replaced by the predicting bounding box. In addition, the weights of neuron z_k will be restored to the previous moment to avoid it still tracking the new target in the next moment. To sum up, if detecting the ID switch, the following operations will be performed:

$$\begin{aligned}\mathbb{B}_k^t &= \hat{\mathbb{B}}_k^{t-1},\\ \hat{\mathbb{B}}^t &= \hat{\mathbb{B}}_k^{t-1} + \widetilde{\boldsymbol{m}}^{t-1},\\ w_{ki}^t &= w_{ki}^{t-1},\end{aligned} \tag{25}$$

where w_{ki} refer to the bias weight and synaptic weights of neuron z_k in the clustering network ($i \geq 0$).

Missed detection: When an object is occluded or moved outside the screen, the neuron z_k of the clustering network may not fire spikes because it cannot find its corresponding pattern, resulting in missed detection of the object. In this case, we use the predicting bounding box to label the missed object. However, if the object has not been detected for a period of time, e.g., 10 timestamps, its corresponding tracker will be assigned to a newly detected object. In summary, when miss the target, we are going to use the predicted results:

$$\begin{aligned}\mathbb{B}_k^t &= \hat{\mathbb{B}}_k^{t-1}\\ \hat{\mathbb{B}}^t &= \hat{\mathbb{B}}_k^{t-1} + \widetilde{\boldsymbol{m}}^{t-1}\end{aligned} \tag{26}$$

IV. Experiments

In this section, we validate the advantages of our proposed SNNTracker by conducting experiments across three types of spike-camera-based MOT datasets, each reflecting different annotation granularities and scene complexity. Specifically, we use: 1) densely annotated datasets with MOT boxes aligned to continuous spike time steps (20 kHz), covering scenarios with camera ego-motion, high-speed deformation, and occlusion; 2) a sparsely annotated dataset adapted from [48], where we associate detection boxes at 50 Hz into MOT tracks to evaluate long-term stability under limited temporal resolution; and 3) several realistic spike sequences under challenging outdoor or low-light conditions, where only qualitative visualizations are provided. A full summary of the datasets is included in the supplementary material.

Our SNNTracker operates in an online and one-pass manner—processing the spike stream frame by frame without access to future information. During the STDP-based online clustering stage, we perform only 5 iterations per time step to update neuron weights. This fixed and lightweight update schedule enables fast adaptation while ensuring convergence to stable tracking representations.

A. Continuous Tracking With GT Evaluation

To compare SNNTracker with prior spike-based MOT methods, SVS and ODTSnet [17], [19], we evaluate the proposed method on two tracking sequences captured by spike cameras, as introduced in ODTSnet [19]: “SRD” and “SRTC.” The SRD sequence features an industrial fan rotating at 2400 RPM with five digits (5 to 9) affixed to its blades, while the SRTC sequence involves a USB fan with three characters (“P,” “K,” and “U”) affixed to its blades, captured as the camera moves from left to right.

Fig. 5 shows the tracking trajectories of the models on these sequences. In this experiment, SVS2 is a modified version of SVS that replaces the original eight-connected region search with our attention-based saccade module. The results indicate that all models successfully capture the trajectories of the rotating digits in the SRD sequence. For the SRTC sequence, which involves both rotational and the camera’s translational ego-motion, both ODTSnet and SNNTracker achieve continuous and accurate trajectory estimation. However, the original SVS algorithm generates numerous erroneous trajectories. Notably, by simply replacing the SVS detection module with our DNF-based attention mechanism, SVS2 outperforms ODTSnet on SRTC, highlighting the limitations of bottom-up spatial clustering in handling background noise. In contrast, the top-down attention mechanism significantly improves detection and enhances tracking performance.

To investigate the impact of blur and continuity on high-speed object tracking, as discussed in Fig. 1, we performed continuous reconstruction of sharp images for the two scenarios with continuous GT tracking annotations. The annotated track IDs were used as bounding box labels and fed into YOLOv5 for training. Validation was conducted on a portion of the annotated data that was excluded from the training set. Fig. 6 presents partial detection and tracking results for all spike-based MOT algorithms, as well as YOLOv5, on the two spike sequences. To objectively compare the tracking performance of the three methods, we computed the multi-object tracking accuracy (MOTA) [81] on these sequences, as shown in Table I.

As depicted in the qualitative results in Figs. 6 and 5, the SNNTracker algorithm achieves the best tracking performance on the camera-fixed SRD sequence, while the SVS algorithm performs the worst. Although SVS exhibits higher multi-target prediction accuracy on the SRD sequence, it suffers from numerous misjudgments, resulting in significantly lower MOTA scores. ODTSnet also produces many false positives and ID switches, with a substantial total number of misclassified and falsely tracked objects. This is primarily due to the sensitivity of the DBSCAN clustering method to parameter settings; noise points between categories often cause ODTSnet to merge two distinct objects into the same category. Furthermore, when the predicted tracking frame deviates significantly, the Kalman filter reallocates the tracker upon re-identifying the object, causing frequent ID switches.

Both SVS2 and YOLOv5 perform well in high-speed multi-object tracking, emphasizing the importance of accurate detection. However, YOLOv5 occasionally mislabels objects in unseen scenes (Fig. 6), reflecting the limitations of supervised offline models. In contrast, SNNTracker performs online tracking with dynamic weight adaptation via WTA, maintaining consistent identity and achieving 100% MOTA.

In the extended SRTC sequence with additional occlusion (“SRTC long”), we annotated 1250 frames to test robustness.

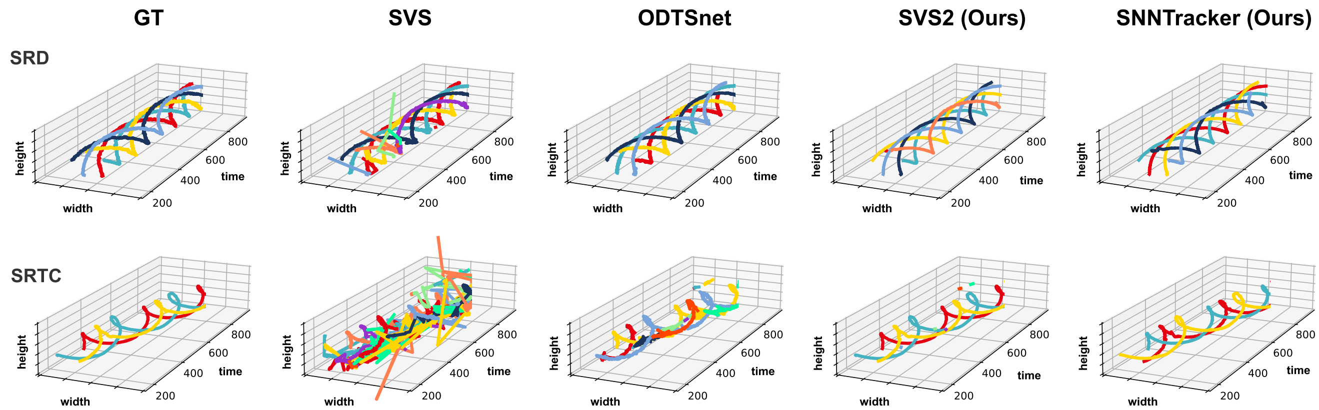

Fig. 5. Comparison of tracking trajectories for different spike-based MOT algorithms on continuous spike datasets (20 kHz).

Fig. 6. Examples of tracking results on the spiking datasets. The tracked bounding boxes are encoded with different colors, and the white boxes with the label "GT" denote the ground truth.

TABLE I
Quantitative Comparison of Multi-Target Tracking Results in Spiking Datasets. Our Proposed Methods Are Highlighted.

Sequence	Method	IDF1 (%)	IDP (%)	IDR (%)	Rcll (%)	Prcn (%)	GT	MT	FP	FN	IDs	FM	MOTA (%)	IDt	IDa	IDm	Training Frames	Test Frames
SRD	SVS	93.10	92.40	93.80	99.90	98.30	5	5	69	5	9	1	97.90	3	3	0	0	799
	ODTSnet	100.00	100.00	100.00	100.00	100.00	5	5	0	0	0	0	100.00	0	0	0	0	799
	SVS2	99.70	99.80	99.50	99.70	100.00	5	5	0	12	1	1	99.70	0	1	0	0	799
	SNNTracker	100.00	100.00	100.00	100.00	100.00	5	5	0	1	0	1	100.00	0	0	0	0	799
	YOLO v5	99.70	99.90	99.80	100.00	99.80	5	5	3	0	2	0	99.60	2	0	0	540	258
SRTC	SVS	23.30	17.70	33.80	97.20	51.00	3	3	2236	67	43	18	2.10	16	13	0	0	799
	ODTSnet	73.80	70.80	77.20	97.20	89.00	3	3	287	68	6	5	84.90	3	2	0	0	799
	SVS2	99.30	98.50	100.00	100.00	98.70	3	3	32	0	0	0	98.70	0	0	0	0	799
	SNNTracker	100.00	100.00	99.90	99.90	100.00	3	3	0	1	0	1	100.00	0	0	0	0	799
	YOLO v5	98.50	99.90	99.80	100.00	97.40	3	3	29	0	4	0	96.90	4	0	0	450	349

TABLE II
PERFORMANCE COMPARISON ON SRTC LONG SEQUENCE

Method	IDF1	IDP	IDR	Rcll	Prcn	FP	FN	IDs	FM	MOTA	#Train	#Test
YOLO v5	95.60	97.60	96.80	97.10	94.80	123	66	10	12	91.40	450	800
SVS2	91.00	90.80	91.20	99.60	99.10	33	15	6	2	98.50	0	1250
SNNTracker	**99.50**	**99.20**	**99.80**	**99.80**	**99.30**	27	7	4	2	**99.00**	0	1250

As shown in Table II, YOLOv5's performance degrades under occlusion, while SVS2 and SNNTracker remain stable, validating the advantage of spike-based online prediction filters in challenging scenarios.

B. Comparison With MOT Using Discrete GT

Since YOLOv5 is inherently a detection-based algorithm, directly assigning tracked IDs as GT labels for detection boxes may not be ideal. To address this, we further evaluate various MOT algorithms on our upgraded discrete spike multi-object tracking dataset (50 Hz) to assess whether these specialized MOT methods can achieve continuous tracking on reconstructed sharp images.

To ensure a fair comparison and avoid performance variations due to hyperparameter tuning, we did not retrain the MOT algorithms. Instead, since all selected MOT methods rely on detection-based tracking, we directly used the GT bounding boxes as input detections. The primary algorithms compared include SORT [20], OCSORT [37], HybridSORT [34], ByteTrack [56], MOTDT [82], and DeepSORT [36]. Among these, MOTDT and DeepSORT require both detection boxes and image feature extraction for association-based tracking. For testing these methods, we employ the state-of-the-art spike-to-sharp image reconstruction algorithm, BSF [45], to recover sharp images from spike data before feeding them into the MOTDT and DeepSORT networks.

The evaluation covers character tracking at five rotational speeds (200–2600 RPM) across six spike sequences: *200R4C*, *800R4C*, *1500R4C*, *2000R4C*, *2600R4C*, *200R5D*, and *800R5D*. These include both four-character (*P, A, M, I*) and five-digit (*1–5*) scenarios, with each sequence spanning four seconds and annotated at 50 Hz (200 GT frames). To illustrate how our SNNTracker performs online clustering, we visualize the learned synaptic weights of output neurons at different time steps in Fig. 9. Each neuron specializes in distinct motion patterns through STDP updates, and its weights align closely with ground truth object positions, even across long sequences. This supports the effectiveness of our unsupervised clustering mechanism in enabling robust, real-time multi-object tracking under varied motion speeds and target configurations.

The qualitative comparison and quantitative MOTA results are shown in Fig. 7 and Table III, respectively. Even when provided with GT bounding boxes and clear, sharp images as input, the MOT algorithms fail to track multiple rapidly rotating objects effectively. In contrast, spike-based methods can robustly process raw spike streams to achieve continuous tracking results.

TABLE III
PERFORMANCE COMPARISON OF MOT METHODS WITH DISCRETE TRACKED GT

Sequence	Method	FP	FN	IDs	MOTA (%)	Sequence	FP	FN	IDs	MOTA (%)
200R4C	SVS	3	2	14	97.60	2600R4C	38	1	74	85.90
	ODTSnet	0	15	76	88.60		1	2	41	94.50
	SVS2	0	0	4	99.50		8	0	0	99.00
	SNNTracker	0	0	4	99.50		1	3	8	98.50
	SORT	0	788	8	0.50		0	507	289	0.50
	OCSORT	0	788	8	0.50		0	396	400	0.50
	HybridSORT	0	788	8	0.50		0	396	400	0.50
	ByteTrack	0	617	57	15.70		0	35	761	0.50
	MOTDT	0	800	0	0.00		144	187	606	-17.10
	DeepSORT	0	800	0	0.00		195	220	570	-23.10
800R4C	SVS	2	0	22	97.00	200R5D	18	1	2	97.90
	ODTSnet	0	12	79	88.60		0	561	139	30.00
	SVS2	0	0	2	99.80		0	17	1	98.20
	SNNTracker	0	0	2	99.80		1	4	2	99.30
	SORT	0	555	241	0.50		0	422	573	0.50
	OCSORT	0	484	312	0.50		0	394	600	0.60
	HybridSORT	0	504	291	0.60		0	420	573	0.70
	ByteTrack	0	121	674	0.50		0	182	810	0.80
	MOTDT	0	391	405	0.50		62	363	629	-5.40
	DeepSORT	209	340	455	-25.50		0	391	603	0.60
1500R4C	SVS	14	7	55	90.50	800R5D	33	2	12	95.30
	ODTSnet	0	4	11	98.10		0	460	276	26.40
	SVS2	0	0	2	99.80		0	54	79	86.70
	SNNTracker	0	0	2	99.80		1	4	25	96.60
	SORT	0	566	226	1.00		0	985	10	0.50
	OCSORT	0	513	283	0.50		0	985	10	0.50
	HybridSORT	0	515	280	0.60		0	985	10	0.50
	ByteTrack	0	115	681	0.50		0	878	33	8.90
	MOTDT	0	712	65	2.90		0	1000	0	0.00
	DeepSORT	118	509	284	-13.90		0	1000	0	0.00
2000R4C	SVS	34	1	81	85.50					
	ODTSnet	0	8	29	95.40					
	SVS2	11	0	0	98.60					
	SNNTracker	1	7	8	98.00					
	SORT	0	788	8	0.50					
	OCSORT	0	788	8	0.50					
	HybridSORT	0	788	8	0.50					
	ByteTrack	0	466	326	1.00					
	MOTDT	0	800	0	0.00					
	DeepSORT	0	800	0	0.00					

In our experiments, we evaluated spike-based tracking under different temporal resolutions. For moderate-speed scenarios (200–800 RPM), sampling every 10 spike steps sufficed, while higher-speed cases (1500–2600 RPM) required denser sampling (every 5 steps) to maintain accuracy. This confirms the importance of leveraging spike stream continuity for robust high-speed tracking. As shown in Fig. 7, SNNTracker consistently maintains stable tracking even with increased object count and extended sequence durations, whereas spike-based SORT shows degraded performance in such dense conditions. Notably, we also verified

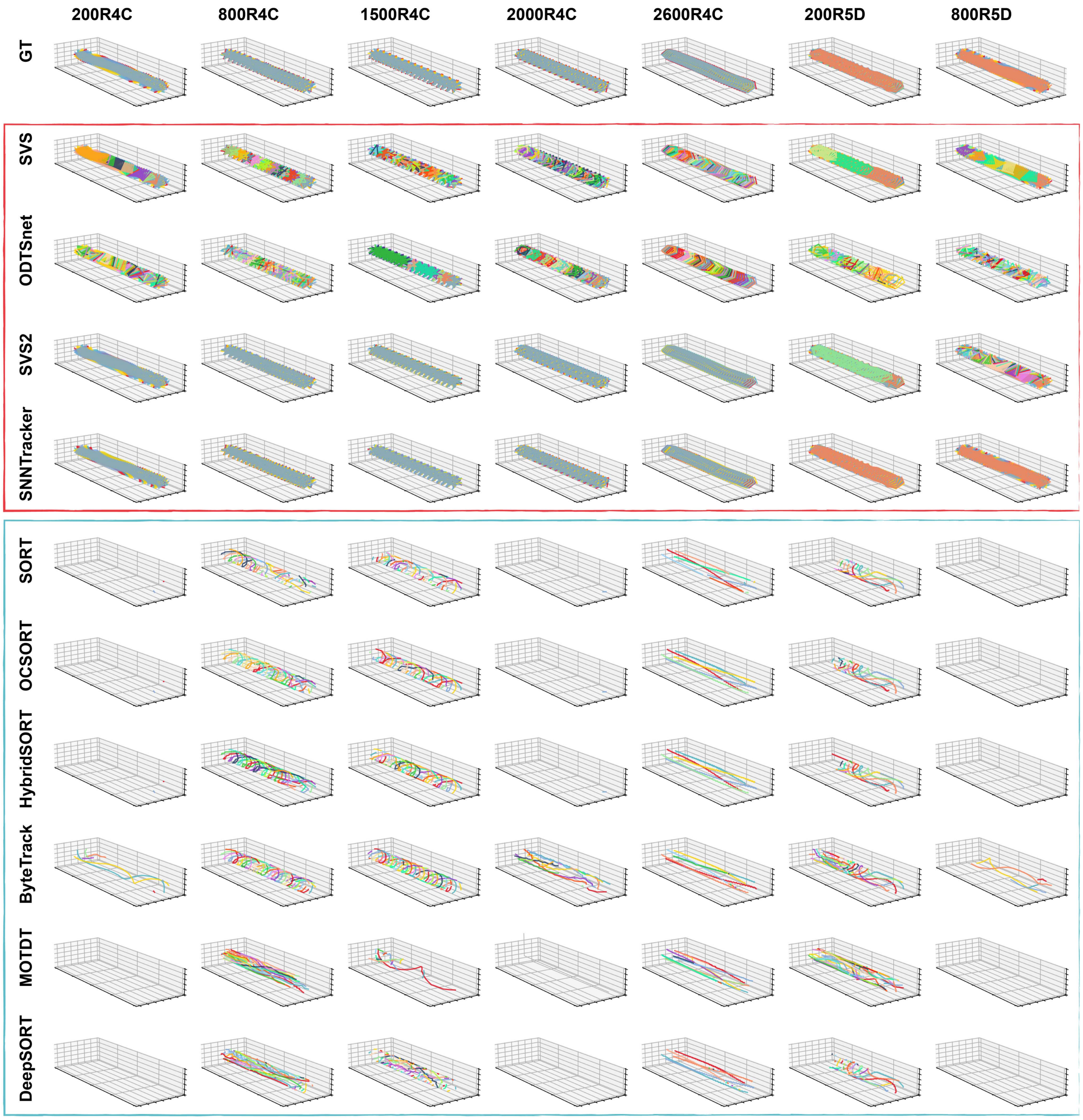

Fig. 7. Comparison of tracking trajectories between spike-based MOT algorithms and image-based MOT algorithms on discrete (50 Hz) tracked GT spike datasets.

that ANN-based MOTA algorithms can achieve strong performance on our SRD and SRTC datasets when using 20 kHz reconstructed images and detection results as input; the corresponding results are included in the supplementary material. Additionally, we manually annotated 13 UAV sequences captured in outdoor environments with complex backgrounds. SNNTracker achieves a MOTA of 97.9% in these sequences, demonstrating its robustness under real-world, low-light conditions. Please refer to the supplementary video and document for details.

C. Ablation Study on Complex Scenes

To further validate the robustness and generalization of our SNNTracker in real-world, unlabeled settings, we conducted additional experiments on several challenging sequences without annotations. These scenarios were designed to test the system's adaptability beyond the controlled experimental datasets. In addition to the previously tested *SRTC long* sequence, we introduced three complex indoor sequences—*SRTC+OCC1*, *SRTC+OCC2*, and *Badminton*—featuring occlusion, rapid ego-motion, and abrupt trajectory changes. Furthermore, we recorded two new outdoor sequences: a basketball playing scene and a vehicle-following scenario under natural illumination and dynamic backgrounds. Despite the lack of manual annotations, visualizations in the supplementary video demonstrate that SNNTracker maintains stable and accurate tracking in these unconstrained environments.

These results highlight the advantages of using a fully spike-driven SNN clustering pipeline. Unlike SVS2, which depends on detection and association heuristics, SNNTracker adapts to

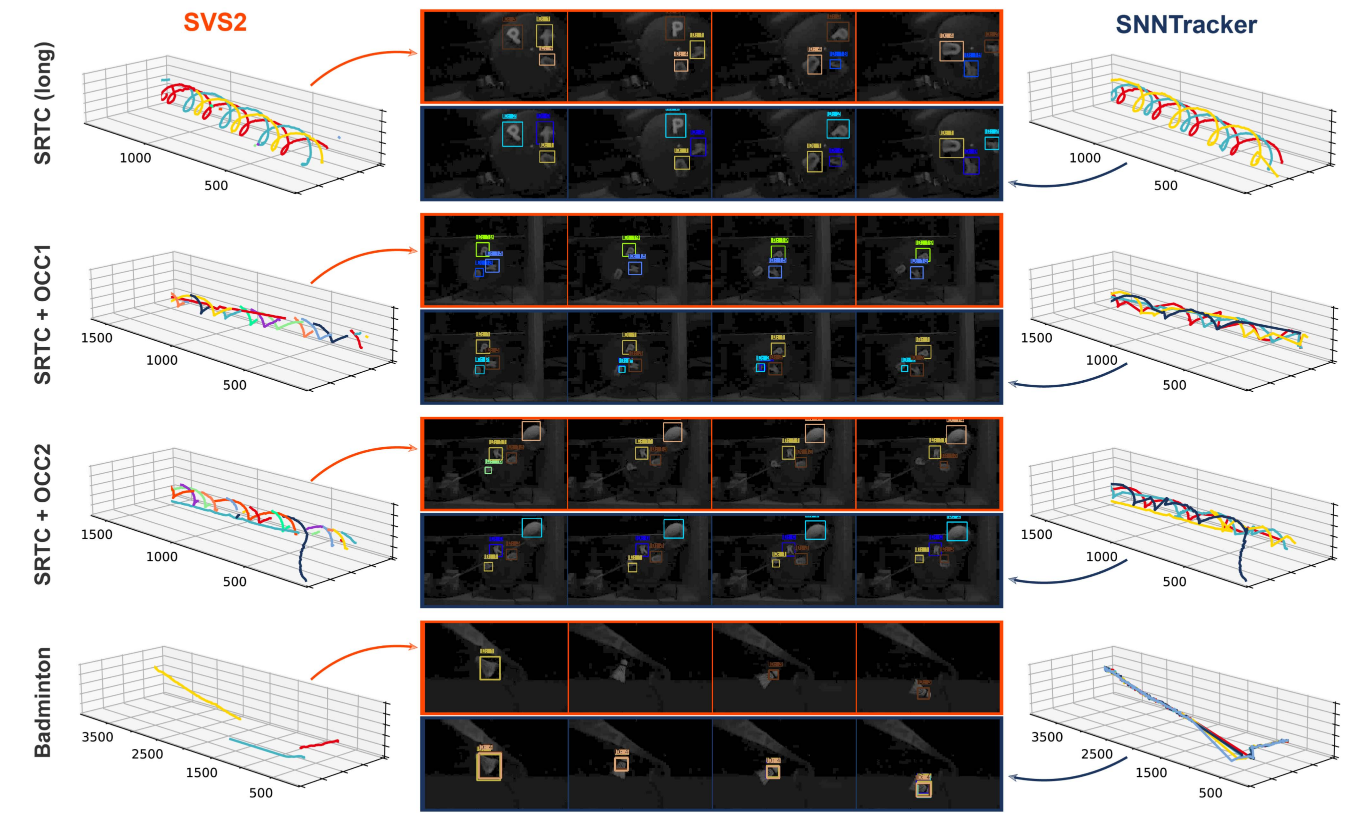

Fig. 8. Comparison of tracking results between SVS2 and SNNTracker on the complex continuous spiking dataset without GT.

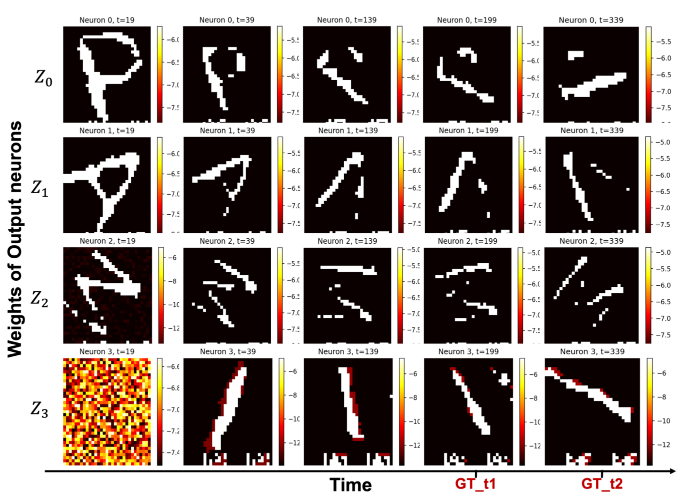

Fig. 9. STDP-learned synaptic weights of output neurons at different times, showing motion specialization and alignment with GT for online clustering.

evolving visual input by leveraging STDP-based online learning and a WTA mechanism. This enables unsupervised neuron specialization for each moving object, even when object shapes deform or occlusions occur. As shown in Fig. 8, the tracking trajectories produced by SNNTracker remain consistent and resilient across diverse scenes, confirming the effectiveness of integrating STDP clustering with motion estimation and attention-guided saccades.

Taken together, these findings reinforce the importance of our proposed fully SNN-based modules—particularly the STDP clustering network—for enabling robust, real-time multi-object tracking in complex and annotation-sparse environments.

Computational Efficiency Analysis and Hardware Potential: All the algorithms in this study were implemented using the PyTorch framework, with all variables represented in tensor form. We evaluated the computational performance and resource usage of four spike-based multi-object tracking algorithms on an NVIDIA RTX 4090 GPU. As shown in Fig. 10, the evaluation focused on four key performance metrics: CPU time, CUDA time, peak memory usage, and the inference time required per spike frame. The results reveal that ODTSnet exhibits significant variability in computational efficiency due to the dependency of its DBSCAN algorithm on the number of filtered clustering points. This variability is especially pronounced in the SRD scenario, where tracking five moving digits requires a sharp increase in both processing time and computational resources. In contrast, when comparing SVS2 and SNNTracker, the SNN-based association and tracking approach consistently demonstrates lower computational time compared to the traditional SORT framework, which relies on the Hungarian algorithm for association and Kalman filtering for trajectory estimation. Among all the evaluated methods, SNNTracker achieved the lowest average tracking time, highlighting its computational efficiency.

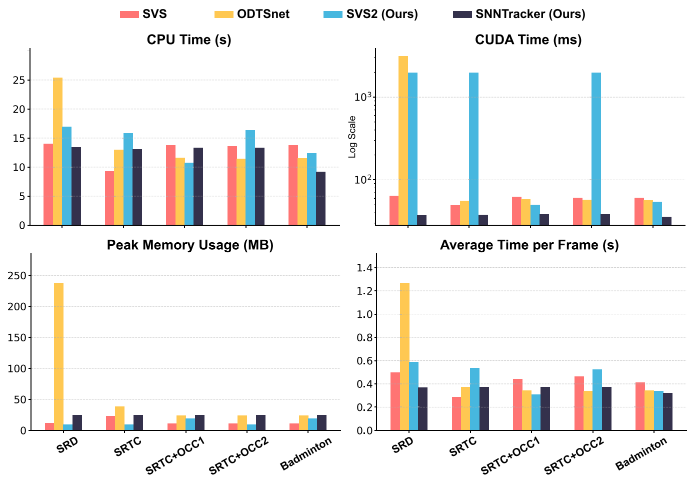

Fig. 10. Comparison of speed and resource usage among different spike-based MOT algorithms.

This finding is particularly exciting when considering the hardware potential. Zhu et al. [83] have previously demonstrated that implementing SVS on an FPGA using a compute-in-memory architecture can boost its processing speed to 20,881 Hz in the SRD scenario while consuming dynamic power of only 1.618 W and static power of 0.636 W. Based on the performance insights in Fig. 10, we believe SNNTracker has significant potential for FPGA hardware implementation, enabling real-time, low-latency, and low-power processing for high-speed multi-object tracking scenarios.

V. Conclusion

In this work, we propose the first fully SNN-based MOT framework that directly processes low-latency, continuous spike streams for robust online tracking. We systematically examine the question: *Why spike?*, and show that spike-based data inherently addresses motion blur and temporal discontinuity, key challenges in traditional image-based tracking. Our experiments reveal that existing tracking algorithms fail under high-speed conditions when relying on sparse annotations or discrete, sharp frames. Even with ground-truth detection boxes, performance deteriorates as motion speed increases. In contrast, when trained with continuous annotations, these methods improve significantly, highlighting the importance of temporal continuity. However, achieving such dense annotations requires costly reconstruction and manual labeling, which limits real-time applicability.

SNNTracker eliminates these limitations by learning directly from raw spike data through online STDP-based clustering. It achieves superior accuracy and robustness across all tested scenarios without any labeled data. Its ability to maintain stable tracking under occlusions, fast motion, and outdoor or low-light conditions demonstrates the power of direct spike processing. Finally, SNNTracker offers strong potential for efficient, low-power deployment on hardware such as FPGAs, enabling real-time applications in autonomous driving, drones, and robotic systems.

References

[1] Q. Wang, L. Zhang, L. Bertinetto, W. Hu, and P. H. Torr, "Fast online object tracking and segmentation: A unifying approach," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 1328–1338.

[2] D. Bradley, P. Bell, O. Landen, J. Kilkenny, and J. Oertel, "Development and characterization of a pair of 30–40 ps X-ray framing cameras," *Rev. Sci. Instrum.*, vol. 66, no. 1, pp. 716–718, 1995.

[3] J. Itatani, F. Quéré, G. L. Yudin, M. Y. Ivanov, F. Krausz, and P. B. Corkum, "Attosecond streak camera," *Phys. Rev. Lett.*, vol. 88, no. 17, 2002, Art. no. 173903.

[4] G. Indiveri and R. Douglas, "Neuromorphic vision sensors," *Science*, vol. 288, no. 5469, pp. 1189–1190, 2000.

[5] L. Patrick, C. Posch, and T. Delbruck, "A 128x 128 120 db 15 μ s latency asynchronous temporal contrast vision sensor," *IEEE J. Solid-state Circuits*, vol. 43, no. 2, pp. 566–576, Feb. 2008.

[6] T. Delbrück, B. Linares-Barranco, E. Culurciello, and C. Posch, "Activity-driven, event-based vision sensors," in *Proc. IEEE Int. Symp. Circuits Syst.*, 2010, pp. 2426–2429.

[7] G. Gallego et al., "Event-based vision: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.,*vol. 44, no. 1, pp. 154–180, Jan. 2020.

[8] F. Barranco, C. Fermuller, and E. Ros, "Real-time clustering and multi-target tracking using event-based sensors," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2018, pp. 5764–5769.

[9] A. Mitrokhin, C. Fermüller, C. Parameshwara, and Y. Aloimonos, "Event-based moving object detection and tracking," in *Proc. IEEE Int. Conf. Intell. Robots Syst.*, 2018, pp. 1–9.

[10] T. Stoffregen and L. Kleeman, "Simultaneous optical flow and segmentation (SOFAs) using dynamic vision sensor," 2018, *arXiv:1805.12326*.

[11] C. M. Parameshwara, S. Li, C. Fermüller, N. J. Sanket, M. S. Evanusa, and Y. Aloimonos, "SpikeMS: Deep spiking neural network for motion segmentation," in *Proc. 2021 IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2021, pp. 3414–3420.

[12] Y. Zhou, G. Gallego, X. Lu, S. Liu, and S. Shen, "Event-based motion segmentation with spatio-temporal graph cuts," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 8, pp. 4868–4880, Aug. 2023.

[13] Y. Deng, H. Chen, H. Liu, and Y. Li, "A voxel graph CNN for object classification with event cameras," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 1172–1181.

[14] D. Gehrig, A. Loquercio, K. G. Derpanis, and D. Scaramuzza, "End-to-end learning of representations for asynchronous event-based data," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 5633–5643.

[15] R. W. Baldwin, R. Liu, M. Almatrafi, V. Asari, and K. Hirakawa, "Time-ordered recent event (TORE) volumes for event cameras," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2519–2532, Feb. 2023.

[16] H. Rebecq, G. Gallego, E. Mueggler, and D. Scaramuzza, "Emvs: Event-based multi-view stereo—3D reconstruction with an event camera in real-time," *Int. J. Comput. Vis.*, vol. 126, no. 12, pp. 1394–1414, 2018.

[17] T. Huang et al., "1000× faster camera and machine vision with ordinary devices," *Engineering*, vol. 25, pp. 110–119, 2023.

[18] Y. Zheng et al., "SpikeCV: Open a continuous computer vision era," *Sci. China Inf. Sci.*, 2025.

[19] Y. Zheng, Z. Yu, S. Wang, and T. Huang, "Spike-based motion estimation for object tracking through bio-inspired unsupervised learning," *IEEE Trans. Image Process.*, vol. 32, pp. 335–349, 2023.

[20] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *Proc. 2016 IEEE Int. Conf. Image Process.*, 2016, pp. 3464–3468.

[21] H. W. Kuhn, "The Hungarian method for the assignment problem," *Nav. Res. Logistics Quart.*, vol. 2, no. 1-2, pp. 83–97, 1955.

[22] X. Li, K. Wang, W. Wang, and Y. Li, "A multiple object tracking method using Kalman filter," in *Proc. IEEE Int. Conf. Inf. Automat.*, 2010, pp. 1862–1866.

[23] Y. Dan and M.-m. Poo, "Spike timing-dependent plasticity of neural circuits," *Neuron*, vol. 44, no. 1, pp. 23–30, 2004.

[24] D. E. Feldman, "The spike-timing dependence of plasticity," *Neuron*, vol. 75, no. 4, pp. 556–571, 2012.

[25] M. Ester et al., "A density-based algorithm for discovering clusters in large spatial databases with noise," *Proc. 2nd Int. Conf. Knowl. Discov. Data Mining*, vol. 96, no. 34, pp. 226–231, 1996.

[26] W. Maass, "Networks of spiking neurons: The third generation of neural network models," *Neural Netw.*, vol. 10, no. 9, pp. 1659–1671, 1997.

[27] S. Thorpe, A. Delorme, and R. V. Rullen, "Spike-based strategies for rapid processing," *Neural Netw.*, vol. 14, no. 6-7, pp. 715–725, 2001.

[28] R. VanRullen, R. Guyonneau, and S. J. Thorpe, "Spike times make sense," *Trends Neurosciences*, vol. 28, no. 1, pp. 1–4, 2005.

[29] R. Gütig, "To spike, or when to spike," *Curr. Opin. Neurobiol.*, vol. 25, pp. 134–139, 2014.

[30] W. Maass, "To spike or not to spike: That is the question," in *Proc. IEEE*, vol. 103, no. 12, pp. 2219–2224, Dec. 2015.

[31] R. Brette, "Philosophy of the spike: Rate-based vs. spike-based theories of the brain," *Front. Syst. Neurosci.*, vol. 9, 2015, Art. no. 140675.

[32] Ultralytics, "YOLOv5: Real-time object detection," 2020. Accessed: Jun. 17, 2024. [Online]. Available: https://github.com/ultralytics/YOLOv5

[33] R. J. Douglas and K. A. Martin, "Neuronal circuits of the neocortex," *Annu. Rev. Neurosci.*, vol. 27, pp. 419–451, 2004.

[34] M. Yang et al., "Hybrid-sort: Weak cues matter for online multi-object tracking," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 7, pp. 6504–6512.

[35] K. Yi et al., "Ucmctrack: Multi-object tracking with uniform camera motion compensation," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 7, pp. 6702–6710.

[36] N. Wojke and A. Bewley, "Deep cosine metric learning for person re-identification," in *Proc. 2018 IEEE Winter Conf. Appl. Comput. Vis.*, 2018, pp. 748–756.

[37] J. Cao, J. Pang, X. Weng, R. Khirodkar, and K. Kitani, "Observation-centric sort: Rethinking sort for robust multi-object tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 9686–9696.

[38] Y. Zhang et al., "Bytetrack: Multi-object tracking by associating every detection box," in *Proc. Eur. Conf. Comput. Vis.* Cham: Springer Nature Switzerland, 2024, pp. 1–21.

[39] J. Redmon, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 779–788.

[40] L. Zhu, S. Dong, T. Huang, and Y. Tian, "A retina-inspired sampling method for visual texture reconstruction," in *Proc. 2019 IEEE Int. Conf. Multimedia Expo.*, 2019, pp. 1432–1437.

[41] L. Zhu, J. Li, X. Wang, T. Huang, and Y. Tian, "Neuspike-net: High speed video reconstruction via bio-inspired neuromorphic cameras," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 2400–2409.

[42] Y. Zheng, L. Zheng, Z. Yu, B. Shi, Y. Tian, and T. Huang, "High-speed image reconstruction through short-term plasticity for spiking cameras," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 6358–6367.

[43] J. Zhang, S. Jia, Z. Yu, and T. Huang, "Learning temporal-ordered representation for spike streams based on discrete wavelet transforms," in *Proc. AAAI Conf. Artif. Intell*, 2023, vol. 37, no. 1, pp. 137–147.

[44] Y. Zheng, L. Zheng, Z. Yu, T. Huang, and S. Wang, "Capture the moment: High-speed imaging with spiking cameras through short-term plasticity," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8127–8142, Jul. 2023.

[45] R. Zhao et al., "Boosting spike camera image reconstruction from a perspective of dealing with spike fluctuations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 24955–24965.

[46] J. Han et al., "Hybrid high dynamic range imaging fusing neuromorphic and conventional images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8553–8565, Jul. 2023.

[47] J. Zhang, K. Chen, S. Chen, Y. Zheng, T. Huang, and Z. Yu, "SpikeGS: 3D Gaussian splatting from spike streams with high-speed camera motion," in *Proc. 32nd ACM Int. Conf. Multimedia*, 2024, pp. 9194–9203.

[48] J. Li, X. Wang, L. Zhu, J. Li, T. Huang, and Y. Tian, "Retinomorphic object detection in asynchronous visual streams," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 1332–1340.

[49] J. Zhao, S. Zhang, Z. Yu, and T. Huang, "Recognizing ultra-high-speed moving objects with bio-inspired spike camera," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 7, pp. 7478–7486.

[50] S. Chen, J. Zhang, Y. Zheng, T. Huang, and Z. Yu, "Enhancing motion deblurring in high-speed scenes with spike streams," in *Proc. Adv. Neural Inf. Process. Syst*, 2023, pp. 70279–70292.

[51] K. Chen et al., "SpikeReveal: Unlocking temporal sequences from real blurry inputs with spike streams," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 37, 2024, pp. 62673–62696.

[52] J. Zhang, S. Chen, Y. Zheng, Z. Yu, and T. Huang, "Spike-guided motion deblurring with unknown modal spatiotemporal alignment," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 25047–25057.

[53] L. Ma, X. Yang, and D. Tao, "Person re-identification over camera networks using multi-task distance metric learning," *IEEE Trans. Image Process.*, vol. 23, no. 8, pp. 3656–3670, Aug. 2014.

[54] W.-S. Zheng, X. Li, T. Xiang, S. Liao, J. Lai, and S. Gong, "Partial person re-identification," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 4678–4686.

[55] T. Meinhardt, A. Kirillov, L. Leal-Taixe, and C. Feichtenhofer, "Trackformer: Multi-object tracking with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 8844–8854.

[56] U. Bhattacharya, T. Mittal, R. Chandra, T. Randhavane, A. Bera, and D. Manocha, "Step: Spatial temporal graph convolutional networks for emotion perception from gaits," in *Proc. AAAI Conf. Artif. Intell.*, 2020, vol. 34, no. 02, pp. 1342–1350.

[57] J. Hyun, M. Kang, D. Wee, and D.-Y. Yeung, "Detection recovery in online multi-object tracking with sparse graph tracker," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2023, pp. 4850–4859.

[58] M. Gehrig, W. Aarents, D. Gehrig, and D. Scaramuzza, "DSEC: A stereo event camera dataset for driving scenarios," *IEEE Robot. Automat. Lett.*, vol. 6, no. 3, pp. 4947–4954, Jul. 2021.

[59] A. Z. Zhu, L. Yuan, K. Chaney, and K. Daniilidis, "Unsupervised event-based learning of optical flow, depth, and egomotion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 989–997.

[60] R. P. Costa, P. J. Sjostrom, and M. C. V. Rossum, "Probabilistic inference of short-term synaptic plasticity in neocortical microcircuits," *Front. Comput. Neurosci.*, vol. 7, 2013, Art. no. 75.

[61] W. Gerstner, W. M. Kistler, R. Naud, and L. Paninski, *Neuronal Dynamics: From Single Neurons to Networks and Models of Cognition*. Cambridge U.K.: Cambridge Univ. Press, 2014.

[62] K. Nakayama and G. H. Silverman, "The aperture problem—I. perception of nonrigidity and motion direction in translating sinusoidal lines," *Vis. Res.*, vol. 28, no. 6, pp. 739–746, 1988.

[63] J. E. Hoffman and B. Subramaniam, "The role of visual attention in saccadic eye movements," *Percep. Psychophys.*, vol. 57, no. 6, pp. 787–795, 1995.

[64] W. X. Schneider and H. Deubel, "Visual attention and saccadic eye movements: Evidence for obligatory and selective spatial coupling," in *Studies in Visual Information Processing*, vol. 6. New York, NY, USA: Elsevier, 1995, pp. 317–324.

[65] A. Vaswani et al., "Attention is all you need," 2017, *arXiv:1706.03762*.

[66] P. Sun et al., "Transtrack: Multiple-object tracking with transformer," 2020, *arXiv:2012.15460*.

[67] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[68] M. A. Giese, *Dynamic Neural Field Theory for Motion Perception*, vol. 469. Berlin, Germany: Springer Science & Business Media, 2012.

[69] S.-i. Amari, "Dynamics of pattern formation in lateral-inhibition type neural fields," *Biol. Cybern.*, vol. 27, no. 2, pp. 77–87, 1977.

[70] G. Ning et al., "Spatially supervised recurrent convolutional neural networks for visual object tracking," in *Proc. 2017 IEEE Int. Symp. Circuits Syst.*, 2017, pp. 1–4.

[71] P. Voigtlaender et al., "MOTs: Multi-object tracking and segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 7942–7951.

[72] P. Bergmann, T. Meinhardt, and L. Leal-Taixe, "Tracking without bells and whistles," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 941–951.

[73] Z. Wang, L. Zheng, Y. Liu, and S. Wang, "Towards real-time multi-object tracking," 2019, *arXiv:1909.12605*.

[74] B. Nessler, M. Pfeiffer, and W. Maass, "Stdp enables spiking neurons to detect hidden causes of their inputs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2009, vol. 22, pp. 1357–1365.

[75] B. Nessler, M. Pfeiffer, L. Buesing, and W. Maass, "Bayesian computation emerges in generic cortical microcircuits through spike-timing-dependent plasticity," *PLoS Comput. Biol.*, vol. 9, no. 4, 2013, Art. no. e1003037.

[76] W. Gerstner and W. M. Kistler, *Spiking Neuron Models: Single Neurons, Populations, Plasticity*. Cambridge U.K.: Cambridge Univ. Press, 2002.

[77] R. P. Rao and D. H. Ballard, "Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects," *Nature Neurosci.*, vol. 2, no. 1, pp. 79–87, 1999.

[78] K. Friston, "Predictive coding, precision and synchrony," *Cogn. Neurosci.*, vol. 3, no. 3-4, pp. 238–239, 2012.

[79] A. Clark, "Whatever next? predictive brains, situated agents, and the future of cognitive science," *Behav. Brain Sci.*, vol. 36, no. 3, pp. 181–204, 2013.

[80] K. Friston, "Does predictive coding have a future?," *Nature Neurosci.*, vol. 21, no. 8, pp. 1019–1021, 2018.

[81] K. Bernardin and R. Stiefelhagen, "Evaluating multiple object tracking performance: The clear mot metrics," *EURASIP J. Image Video Process.*, vol. 2008, pp. 1–10, 2008.

[82] C. Long, A. Haizhou, Z. Zijie, and S. Chong, "Real-time multiple people tracking with deeply learned candidate selection and person re-identification," in *Proc. IEEE Int. Conf. Multimedia Expo.*, 2018, pp. 1–6.

[83] Y. Zhu, Y. Zhang, X. Xie, and T. Huang, "An fpga accelerator for high-speed moving objects detection and tracking with a spike camera," *Neural Comput.*, vol. 34, no. 8, pp. 1812–1839, 2022.

Yajing Zheng (Associate Member) received the BS degree from Sichuan University, Sichuan, China, in 2017, and the PhD degree from the School of Computer Science, Peking University, Beijing, China, in 2022. She is currently a boya postdoctoral fellow with Peking University. Her research interests include neuroscience, brain-inspired computing, machine learning, spiking neural networks, neuromorphic computing, and computational neuroscience.

Chengen Li received the BS degree from the School of Computer Science, Chongqing University, Chongqing, China, in 2022. He is currently working toward the MEng degree with the School of Software and Microelectronics, Peking University, Beijing, China. His research interests include visual information processing and multi-modality learning.

Jiyuan Zhang (Member, IEEE) received the BE degree from College of Computer Science and Technology, Zhejiang University, Zhejiang, China, in 2020. He is currently working toward the PhD degree with the School of Computer Science, Peking University, Beijing, China. His current research interests include neuromorphic computing, computer vision, and computational photography.

Zhaofei Yu (Member, IEEE) received the BS degree from the Hong Shen Honors School, College of Optoelectronic Engineering, Chongqing University, Chongqing, China, in 2012 and the PhD degree from the Automation Department, Tsinghua University, Beijing, China, in 2017. He is currently an assistant professor with the Institute for Artificial Intelligence, Peking University, Beijing. His current research interests include artificial intelligence, brain-inspired computing, and computational neuroscience.

Tiejun Huang (Senior Member, IEEE) received the bachelor's and master's degree in computer science from the Wuhan University of Technology, Wuhan, in 1992, and 1995, respectively and the PhD degree in pattern recognition and intelligent systems from Huazhong (Central China) University of Science and Technology, Wuhan, China, in 1998. He is currently a professor with the School of Computer Science, Peking University, Beijing, China, and the dean of Beijing Academy of Artificial Intelligence (BAAI). He has authored or coauthored more than 100 peer-reviewed papers and three books. His research interests include video coding, image understanding, digital right management, and digital library. He was the recipient of the National Science Fund for Distinguished Young Scholars, Changjiang Distinguished Professorship, and National Talent Program for Technological Innovation Leadership.